

NYANDOMA, Russian Federation

WMO: 228540

Lat: 61.6794N

Lon: 40.1953E

Elev: 233

StdP: 98.56

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -28.9	(c) -25.5	(d) -32.0	(e) 0.2	(f) -28.9	(g) -28.2	(h) 0.3	(i) -25.4	(j) 7.6	(k) -5.4	(l) 7.0	(m) -4.8	(n) 1.7	(o) 20	(p) 0.637

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.8	26.6	18.3	24.9	17.4	23.1	16.3	19.5	24.6	18.5	23.2	17.5	21.6	2.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	13.0	21.9	16.7	12.2	20.7	15.6	11.4	19.8	56.4	24.7	53.4	23.4	50.0	21.3	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.8	6.1	5.5	-30.2	29.5	4.0	1.6	-33.1	30.7	-35.5	31.7	-37.7	32.6	-40.7	33.8
			-30.3	21.1	4.0	1.3	-33.2	22.1	-35.5	22.9	-37.8	23.6	-40.7	24.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.1	-11.8	-9.9	-5.3	1.8	8.4	13.5	16.5	13.8	8.7	1.9	-4.9	-8.8		
	DBStd	11.10	7.86	7.16	5.00	5.17	5.50	4.54	3.70	3.70	3.83	4.32	6.09	6.90		
	HDD10.0	3430	674	556	476	250	99	16	2	8	68	252	447	583		
	HDD18.3	5977	933	789	734	495	311	156	79	147	288	508	697	841		
	CDD10.0	538	0	0	0	5	49	121	205	126	30	2	0	0		
	CDD18.3	44	0	0	0	0	2	12	24	7	0	0	0	0		
	CDH23.3	338	0	0	0	1	19	107	164	46	0	0	0	0		
	CDH26.7	48	0	0	0	0	1	21	22	4	0	0	0	0		
(14) Wind	WSAvg	2.7	2.7	3.0	3.0	2.7	2.7	2.5	2.3	2.3	2.5	2.9	2.9	2.8		
(15) Precipitation	PrecAvg	755	54	46	46	40	63	72	71	80	67	74	75	67		
	PrecMax	916	91	104	76	83	134	132	170	162	142	129	140	118		
	PrecMin	604	17	12	7	8	21	15	21	46	22	20	22	29		
	PrecStd	82	17	22	17	18	25	31	31	32	30	29	25	24		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.7	2.0	7.1	20.6	26.0	29.1	29.2	27.0	21.6	14.3	5.9	2.7		
		MCWB	0.1	1.3	2.3	11.3	15.3	19.2	20.2	18.2	15.9	10.7	5.2	2.2		
	2%	DB	0.2	0.8	4.2	16.0	22.8	26.2	26.8	24.5	18.6	11.0	4.6	1.1		
		MCWB	-0.3	0.2	0.9	9.2	13.8	17.8	18.7	17.4	14.5	9.7	4.1	0.5		
	5%	DB	-0.4	0.0	2.4	12.1	20.3	23.9	25.3	22.3	16.3	9.4	3.5	0.3		
		MCWB	-0.8	-0.5	0.3	6.9	12.4	16.3	18.1	16.4	13.1	8.5	3.1	-0.1		
	10%	DB	-1.5	-0.8	1.2	9.3	17.8	21.7	23.5	20.4	14.5	7.9	2.1	-0.3		
		MCWB	-1.9	-1.3	-0.2	5.0	11.4	15.0	17.1	15.7	12.1	7.0	1.7	-0.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.5	1.5	2.8	12.2	17.0	20.4	21.3	19.6	16.7	11.5	5.5	2.4		
		MCDB	0.6	1.9	5.2	19.7	23.7	26.7	27.3	24.6	20.1	13.2	5.8	2.7		
	2%	WB	-0.1	0.3	1.7	9.5	15.3	19.0	19.8	18.3	15.1	9.7	4.2	0.8		
		MCDB	0.1	0.6	3.4	15.2	20.4	25.1	24.6	22.7	17.9	10.6	4.5	1.0		
	5%	WB	-0.8	-0.4	0.8	7.3	13.4	17.5	19.0	17.3	13.8	8.6	3.1	0.0		
		MCDB	-0.4	-0.1	1.9	11.4	17.9	22.2	23.7	21.1	15.8	9.5	3.4	0.3		
	10%	WB	-1.9	-1.3	0.0	5.4	12.1	16.0	18.1	16.2	12.4	7.1	1.7	-0.7		
		MCDB	-1.5	-0.8	1.0	8.6	17.2	20.4	22.4	19.5	14.1	7.9	2.0	-0.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.4	5.8	7.3	8.6	9.3	9.1	8.8	8.0	6.5	3.9	3.9	4.5		
		MCDBR	4.6	4.6	7.7	12.5	13.2	12.4	11.4	11.0	9.5	5.8	3.6	3.8		
	5% WB	MCWBR	4.6	4.6	5.5	7.1	6.3	5.8	5.0	5.0	5.4	4.5	3.6	3.9		
		MCWBR	4.9	4.1	5.8	11.6	11.2	10.8	10.0	9.5	8.5	5.5	3.5	3.9		
(40) Clear-Sky Solar Irradiance	taub		0.243	0.293	0.319	0.358	0.348	0.347	0.382	0.364	0.333	0.314	0.265	0.213		
	taud		2.012	2.074	2.106	2.216	2.448	2.472	2.384	2.412	2.488	2.387	2.116	1.946		
	Ebn at Noon		437	649	778	825	874	880	833	818	783	650	409	287		
	Edh at Noon		39	74	103	114	98	98	104	94	72	56	35	23		
(44) All-Sky Solar Radiation	RadAvg		0.25	0.84	2.10	3.58	4.89	5.23	5.16	3.70	2.09	0.82	0.23	0.11		
	RadStd		0.04	0.22	0.34	0.38	0.36	0.48	0.45	0.35	0.24	0.10	0.04	0.02		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	+0.65	N/A	N/A	N/A	N/A	N/A	-198	-217	N/A	N/A

Nomenclature: See separate page